

Ismail Aissaoui
Email: i.aissaoui@esi-sba.dz
Phone: +213 660 70 77 96
Portfolio: ismail-aissaoui.vercel.app

Tobias Isenberg (Inria AVIZ)
Maria Jesus Lobo-Gunther (IGN GeoVis)

Subject: Application for PhD Position – Visualization of Plausibility and Bias for Data Resources used in Geographic Digital Twins (PC5)

Dear Tobias Isenberg and Maria Jesus Lobo-Gunther,

I am applying with great enthusiasm for the PC5 PhD position on visualizing plausibility and bias in geographic digital twins, working within Inria's AVIZ team and IGN's GeoVis team. As an engineer specializing in Explainable AI (XAI) and interactive data visualization, I am fascinated by the challenge of making deep learning models transparent, interpretable, and visually intuitive.

A significant focus of my recent work involves ensuring that complex, automated models are trustworthy. By implementing Explainable AI pipelines on high-dimensional data, I have developed techniques to extract and visualize the underlying biases and decision logic of neural architectures. Combining this with my hands-on experience in building performant 3D web dashboards, I am fully equipped to automate the classification of spatial data plausibility. I am eager to research novel ways to depict data imprecision interactively, bridging the gap between raw algorithmic bias detection and human-centric visual analysis. Geographic digital twins rely heavily on varied and often contradictory data sources; creating visual metaphors to express this uncertainty is a compelling problem I am prepared to solve.

I am particularly drawn to this position because it bridges rigorous visualization research with practical mapping challenges at IGN. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I would welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui